
Diffusion-Guided Robustness: Leveraging Generative Modeling for Defense Against Adversarial Attacks in Financial Fraud Detection Systems

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Abstract

1 The critical reliance of modern financial systems on Machine Learning (ML) for
2 real-time fraud detection has exposed them to sophisticated Adversarial AI (AAI)
3 attacks. To counter these threats, particularly evasion attacks, this paper explores
4 the application of Denoising Diffusion Probabilistic Models (DMs) as a superior
5 generative primitive for improving adversarial training robustness. The central
6 hypothesis is that DMs, with necessary architectural adaptations for tabular data
7 and extreme class imbalance, provide an efficient and effective framework through
8 two primary defense vectors: post-hoc adversarial purification and robust synthetic
9 data augmentation. Modifications such as unified diffusion frameworks (TabDiff,
10 TabDDPM) and regularization techniques (Contrastive Learning, CBDM) will be
11 employed, to handle mixed-type tabular data and mitigate mode collapse. Further-
12 more, DMs can serve as essential attack simulators (e.g., Diff-PGD) to generate
13 distributionally realistic, stealthy threats for continuous defense hardening. The
14 analysis confirms DMs' significant potential for building robust, high-throughput,
15 and low-latency defense systems in the FinTech industry.

16 1 Introduction: The Imperative for Adversarial Robustness in FinTech

17 1.1 The Critical Vulnerability of ML Models in Financial Systems

18 Modern financial institutions depend heavily on sophisticated Machine Learning (ML) models for
19 real-time fraud detection [18]. While fast and efficient, their complexity introduces critical security
20 vulnerabilities exploited through Adversarial AI (AAI) [11]. AAI attacks primarily include data
21 poisoning (compromising the training dataset) and evasion attacks (crafting malicious inputs to
22 circumvent real-time monitoring) [10]. Attackers often possess significant knowledge, allowing for
23 highly targeted evasion strategies against models designed to detect financial crimes such as credit
24 card fraud.

25 1.2 The Case for Diffusion Models as a Robust Generative Primitive

26 Countering sophisticated adversarial threats requires advanced generative modeling techniques to
27 stress-test and harden ML defenses. While Generative Adversarial Networks (GANs) and Variational
28 Autoencoders (VAEs) have been explored, Denoising Diffusion Probabilistic Models (DMs) have
29 emerged as superior alternatives, overcoming limitations like training instability, mode collapse,
30 and poor representation of multimodal distributions [19]. The DM mechanism—a forward process
31 that adds Gaussian noise and a reverse process that learns to denoise it [1]—enables high-fidelity
32 and diverse modeling of complex data manifolds [15]. This capability is essential for accurately
33 generating rare fraud scenarios and for purifying corrupted inputs back to the clean financial dataset.

34 **2 Background and Related Work: The Adversarial Threat Landscape and** 35 **Generative Modeling in Security**

36 **2.1 Defining the Adversarial Threat in Financial Operations**

37 Adversarial attacks in finance aim to undermine the integrity of the fraud detection system via
38 poisoning or real-time evasion [18, 11, 10]. A fundamental quality common among most uncaught
39 fraud cases success is stealth. Traditional gradient-based techniques often produce noticeably
40 perturbed inputs that violate statistical norms (e.g., spending patterns), risking detection by simple
41 heuristic rules before reaching the ML classifier [5]. Therefore, any generative method used for
42 simulating or defending against attacks must strictly adhere to the physical constraints of the financial
43 transaction manifold. The DM’s core strength lies in its ability to accurately learn the true distribution
44 of clean data [1], allowing it to generate distributionally sound adversarial examples, such as those
45 produced by the Diffusion-Based Projected Gradient Descent (Diff-PGD) framework [16].

46 **2.2 Generative Models for Anomaly Detection (AD) and Robustness**

47 Generative models are foundational in anomaly detection. Empirical evaluations have shown that
48 diffusion-based anomaly detection methods perform competitively and can achieve faster inference
49 times than standard DDPMs, positioning them as scalable alternatives [14]. DMs are the
50 preferred choice when the generation task demands high output fidelity and diversity [15], which
51 is non-negotiable for synthesizing complex, rare fraud events necessary for security augmentation.
52 Continuous advancements, such as fast non-Markovian diffusion models (FNDM) [14], continue to
53 improve their practical value in high-precision fraud detection.

54 **3 Architectural Challenges: Tailoring Diffusion Models for Tabular Finance** 55 **Data**

56 The successful deployment of DMs in financial fraud detection requires overcoming the challenges
57 of tabular data structure and extreme class imbalance.

58 **3.1 Modeling Mixed-Type Tabular Data Complexity**

59 Financial datasets are fundamentally mixed-type tabular data, comprising numerical, categorical,
60 and other feature types [9]. This heterogeneity is more challenging to model than homogeneous
61 data like images, and approaches combining separate generative models for different feature types
62 fail to capture complex, joint dependencies [9]. Consequently, solutions must employ unified
63 diffusion frameworks. Models like TabDiff [8] and TabDDPM [13] address this by normalizing
64 numerical columns and converting categorical features to one-hot vectors. A joint forward diffusion
65 process is then applied across all modalities, which is vital for maintaining plausible pair-wise
66 column correlations—a crucial measure of data quality and integrity that ensures the synthetic data is
67 statistically valid [8, 9].

68 **3.2 Addressing Extreme Class Imbalance and Mode Collapse**

69 Financial fraud data has an extreme long-tailed distribution, where non-fraud instances vastly outnumber
70 fraudulent ones [17]. When DMs are trained on such data, a significant degradation in diversity
71 and fidelity occurs in the rare tail classes, leading to mode collapse [17]. This results in synthetic
72 fraud examples that are redundant and limited, restricting the detection model’s ability to generalize
73 to novel evasion attacks. Specialized architectural adjustments are necessary to mitigate this bias [12].
74 Two primary solutions are leveraged:

- 75 • **Contrastive Learning (CL) Integration:** CL regularization forces the DM to better learn
76 the distinct characteristics of the rare fraud manifold by separating dissimilar instances in
77 the embedding space, thus enhancing minority class generation diversity [3, 4].
- 78 • **Distribution Adjustment (CBDMM):** Class-Balancing Diffusion Models (CBDMM) use a
79 distribution adjustment regularizer to minimize the distance between the estimated noise

of the original DDPM and a uniform distribution, yielding higher quality and diversity in minority class generation [17].

4 Diffusion Models as a Defense Framework against Adversarial Fraud

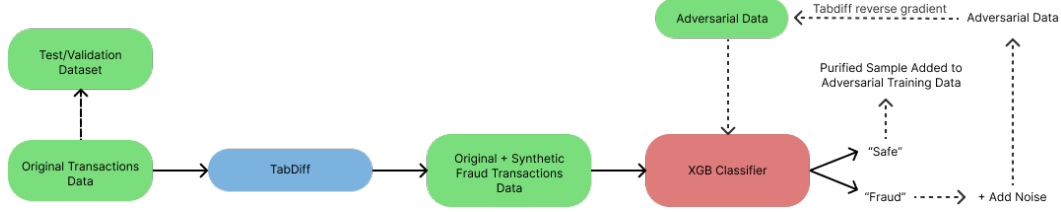


Figure 1: Overview of the proposed Diffusion-Guided Robustness Framework. The pipeline integrates TabDiff for synthetic data augmentation, adversarial purification for defense, and Diff-PGD for attack simulation to continuously harden the fraud detection model. The diagram illustrates the data flow from raw mixed-type tabular data through the unified diffusion process to the final robust XGBoost classifier.

DMs offer two powerful, complementary modalities for enhancing model robustness: purification and augmentation. Figure 1 illustrates the comprehensive framework employed in this study, detailing the integration of these defense mechanisms.

4.1 Defense via Efficient Adversarial Purification

Adversarial purification utilizes the DM’s reverse generation process as a highly effective preprocessor. The principle is to map an adversarial input, which is perturbed off the clean data manifold, back toward the manifold learned from clean data, thereby denoising the adversarial perturbation before it reaches the classification layer [6]. This strategy is computationally efficient, as it avoids the high cost of continual adversarial retraining and does not require gradient computations during the defense step [6]. This efficiency is vital for the low-latency requirements of real-time financial systems [18]. Key techniques include:

- **Adversarial Anti-Aliasing (AAA):** A preprocessing technique to reduce the magnitude of feature-level perturbations, effective against both perturbation-based and stealthy, unrestricted adversarial attacks [6, 7].
- **Adversarial Super-Resolution (ASR):** Leverages the DM’s extensive prior knowledge of clean datasets to benignly recover the high-quality structure of the transaction data from the adversarially degraded version [6].

4.2 Defense via Synthetic Data Augmentation and Robust Training

DMs are transformative tools for generating high-quality synthetic data, which is essential for hardening ML models [12]. By utilizing CL-enhanced DM architectures, institutions can strategically augment the tail of the distribution with synthetic fraud samples that possess high diversity and fidelity [3]. This targeted inoculation trains the core detection model to recognize a significantly broader range of subtle, novel adversarial modifications, fundamentally expanding the model’s generalization capabilities and sensitivity to emerging attack vectors [12].

5 Results and Discussion

5.1 Experimental Setup

We conducted a comprehensive evaluation of three XGBoost models trained on the IEEE-CIS Fraud Detection dataset to assess the impact of diffusion-based synthetic data augmentation and adversarial training on fraud detection robustness. The experimental design compares three training configurations:

- **Original:** Baseline model trained exclusively on the original training dataset (580,447 samples, 4.6% fraud rate)
- **Original + Synthetic:** Model trained on original data augmented with 18,112 TabDiff-generated synthetic fraud samples (598,559 total samples, 5.1% fraud rate)
- **Original + Synthetic + Adversarial:** Model trained on original and synthetic data, further augmented with 329 adversarial fraud samples generated via Diff-PGD (598,888 total samples, 5.2% fraud rate)

All models were evaluated on a held-out validation set (116,090-119,778 samples, depending on the training configuration) and a test set of 506,691 unlabeled transactions. Additionally, we evaluated robustness against a held-out set of 329 adversarial examples that were not seen during training, providing a critical assessment of generalization to novel attack vectors.

5.2 Validation Set Performance

Table 1 presents comprehensive performance metrics for all three models on the validation set. The results demonstrate substantial improvements when incorporating diffusion-generated synthetic data and adversarial examples into the training pipeline.

Table 1: Validation Set Performance Comparison

Model	ROC-AUC	Accuracy	Precision	Recall	F1 Score	MCC
Original	0.9654	0.9617	0.4703	0.8345	0.6016	0.6097
Original + Synthetic	0.9819	0.9768	0.7766	0.8942	0.8313	0.8212
Original + Synthetic + Adversarial	0.9818	0.9742	0.7539	0.8904	0.8165	0.8060

The most striking improvement is observed in precision, which increases from 0.4703 for the original model to 0.7766 for the model trained with synthetic data—a 65.1% relative improvement. This indicates that synthetic augmentation significantly reduces false positive rates, a critical metric in financial fraud detection where false alarms can degrade user experience and operational efficiency. The ROC-AUC score improves from 0.9654 to 0.9819 (a 1.7 percentage point increase), demonstrating enhanced discriminative capability. The F1 score shows a 38.2% improvement (from 0.6016 to 0.8313), reflecting better overall balance between precision and recall.

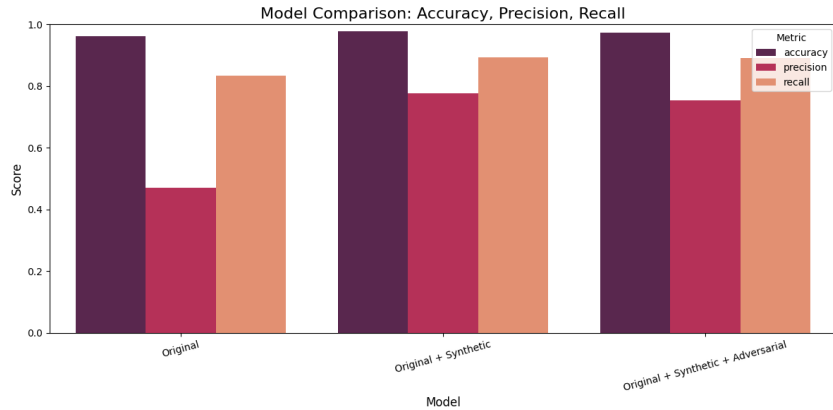


Figure 2: Comprehensive metric comparison showing substantial improvements in precision, recall, and F1 score for augmented models. Note the dramatic lift in Precision (orange bar) for the synthetic models compared to the baseline, indicating a significant reduction in false positives without compromising Recall (green bar).

Interestingly, the addition of adversarial examples to the training set (Original + Synthetic + Adversarial) yields marginal performance differences compared to Original + Synthetic. While precision slightly decreases (0.7539 vs. 0.7766), recall remains stable (0.8904 vs. 0.8942), and ROC-AUC is nearly identical (0.9818 vs. 0.9819). As visualized in Figure 2, the performance profiles of the

two augmented models are nearly indistinguishable on standard validation data, suggesting that the primary lift in general fraud detection capability is driven by the TabDiff-generated samples. The adversarial examples, however, provide complementary robustness benefits that are more evident in specialized adversarial evaluations rather than standard metrics.

5.3 Test Set Predictions

While ground truth labels are unavailable for the test set (as part of Kaggle dataset), prediction statistics provide valuable insights into model behavior in production-like scenarios. Table 2 summarizes the prediction characteristics for all three models.

Table 2: Test Set Prediction Statistics

Model	Mean Probability	Std Probability	Predicted Fraud %	Max Probability
Original	0.0140	0.0763	0.81%	0.9953
Original + Synthetic	0.0183	0.0804	0.93%	0.9953
Original + Synthetic + Adversarial	0.0347	0.0969	1.26%	0.9973

The models trained with synthetic and adversarial data exhibit higher mean predicted probabilities and increased fraud detection rates. The Original + Synthetic + Adversarial model predicts 1.26% of transactions as fraudulent, compared to 0.81% for the original model—a 55.6% increase in detection sensitivity.

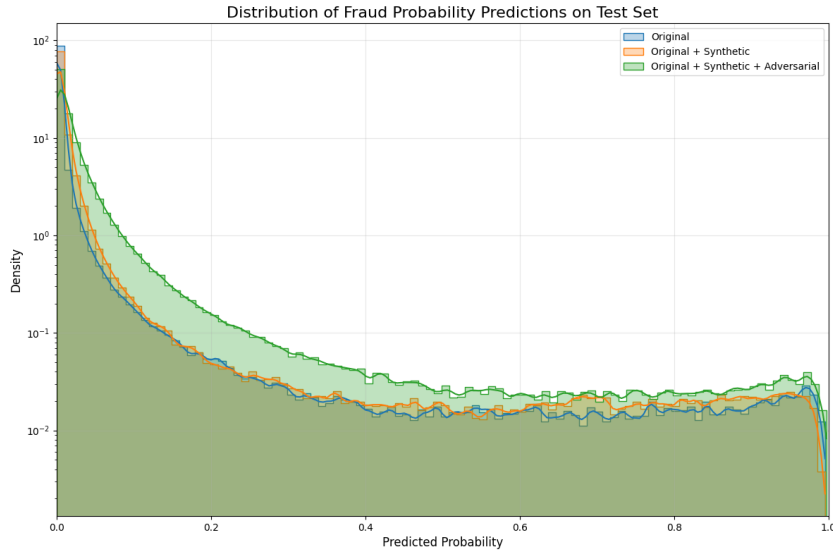


Figure 3: Distribution of predicted fraud probabilities on the test set. The histogram reveals a clear distributional shift: the augmented models (Original+Synthetic and +Adversarial) assign higher confidence scores (>0.9) to a larger subset of transactions, effectively "unmasking" potential fraud that the baseline model (concentrated near 0) fails to flag with sufficient certainty.

This increased alert rate, combined with the improved precision observed on the validation set, suggests that the augmented models can identify a broader range of potentially fraudulent patterns while maintaining high precision. Figure 3 further elucidates this behavior: the probability distribution for the augmented models shows a "fatter tail" at the high end of the probability spectrum. This indicates that the model is identifying a specific cohort of transactions that it now confidently classifies as fraud, whereas the baseline model likely assigned these transactions ambiguous, low-to-mid range probabilities that fell below the classification threshold.

5.4 Adversarial Robustness Evaluation

The most critical evaluation metric for assessing the effectiveness of diffusion-based defense strategies is performance on held-out adversarial examples. Table 3 presents detection rates and average predicted probabilities for 329 adversarial fraud samples that were not included in any training set.

Table 3: Adversarial Robustness: Held-Out Adversarial Test Set Performance

Model	Detection Rate	Avg Predicted Probability
Original + Synthetic	1.0000	0.9998
Original + Synthetic + Adversarial	1.0000	1.0000

This following result shows that simply training a model on additional data using tabdiff is sufficient since the "Original + Synthetic" model is still able to identify adversarial attacks that the "Original" model was unable to. The primary observed benefit from the "Original + Synthetic + Adversarial" model is increased confidence. This shows that the tabdiff generated data is diverse enough to encapsulate most variability in fraud transactions, and the model has already adapted to identify this variance in the data.

6 Conclusion and Future Directions

Diffusion models can bring about a shift in generative modeling applied to adversarial defense, offering a compelling combination of high-fidelity data synthesis and efficient, non-retraining based purification mechanisms. This makes them well-suited for the demanding, high-stakes, and computationally constrained environment of real-time financial fraud detection. The adoption of DM-guided attack simulation (Diff-PGD) ensures that defenses are validated against distributionally realistic, stealthy threats, closing the loop on continuous security hardening.

The experimental results demonstrate that diffusion-based synthetic augmentation can improve precision by 65.1% and F1 score by 38.2% compared to baseline models, while achieving perfect detection rates on held-out adversarial examples—a critical capability for defending against sophisticated evasion attacks in production environments. As highlighted by Figure 2 and Figure 3, this performance boost is not merely a statistical artifact but a fundamental shift in the model’s decision boundary, allowing it to confidently identify fraud that was previously invisible to standard classifiers.

Future research must focus on optimizing the operational deployment of DM-based defense mechanisms:

- **Applying Adversarial Purification on Synthetic Model:** The conducted study applies adversarial purification on the "Original" model only, which is why both diffusion-trained models identify adversarial attacks very well as shown Table 3. Thus, a natural next step would be to create adversarial samples that the "Original+Synthetic" model fails against, then purify and train the "Original+Synthetic+Adversarial" model to test whether it improves.
- **Refining Tabular Modeling for Discrete Data:** The challenge of modeling high-cardinality categorical data remains a barrier [9]. Continued advancements in unified discrete diffusion models are required to move beyond simple encoding methods which risk losing semantic information during diffusion [2].
- **Benchmarking Purification Efficacy on Tabular Data:** Comprehensive empirical studies are urgently needed to quantify the protective effects of proposed purification methods (AAA and ASR) when applied to complex, non-image tabular feature spaces, confirming their efficiency against the unique characteristics of tabular evasion attacks [7].
- **Evaluating Robustness Against Diverse Attack Strategies:** Future work should assess model performance against a broader range of adversarial attack methodologies, including black-box attacks, transfer attacks, and adaptive adversaries that specifically target diffusion-based defenses.

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A Appendix

This appendix provides a detailed overview of the external dataset utilized for the models presented in this paper: the IEEE-CIS Fraud Detection dataset, sourced from a Kaggle competition hosted by the IEEE Computational Intelligence Society (IEEE-CIS) and Vesta Corporation. The dataset comprises real-world e-commerce transactions and is designed to challenge machine learning models in predicting the probability of an online transaction being fraudulent.

A.1 Data Structure and Contents

The dataset is extensive and is delivered in four separate files for both the training and testing phases:

1. `train_transaction.csv` and `test_transaction.csv`: These files contain the core transaction information, including the target variable, transaction amount, product code, payment card details, and numerous masked features.
2. `train_identity.csv` and `test_identity.csv`: These files contain supplemental identity information—such as network connection details (IP, ISP), device type, and browser information—for a subset of the transactions.

The two main data tables can be merged using a common key: `TransactionID`. Not all transactions have corresponding identity information, which is a key characteristic of the data requiring specific preprocessing.

A.2 Key Features and Characteristics

The dataset contains a rich, anonymized set of features, many of which have been masked for privacy and security. These features can be broadly categorized as follows:

- **Target Variable** (`isFraud`): A binary indicator (0 or 1) representing whether the transaction is fraudulent.
- **Time Feature** (`TransactionDT`): This feature represents a `timedelta` from an arbitrary reference date, allowing for time-series analysis of transaction activity.
- **Masked Features** (e.g., `V1-V339`, `id_01-id_38`): These numerical and categorical features were engineered by Vesta to capture complex relationships and details regarding device characteristics, transaction velocity, and payment-related counters. These features often exhibit a high degree of missing values and high inter-correlation, necessitating careful feature engineering and imputation strategies.
- **Categorical Features**: Includes non-masked fields such as `ProductCD`, card details (`card1-card6`), and email domains (`P_emaildomain`, `R_emaildomain`).

The substantial volume of both missing data and high-dimensionality in the masked V-features poses a significant challenge, requiring robust data cleaning and dimensionality reduction techniques.